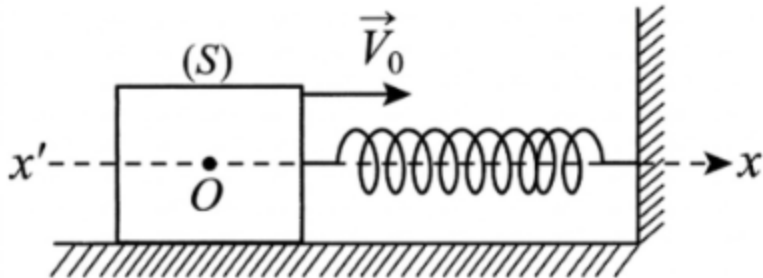


Mechanical Oscillations Worksheet (GS + LS)

Ex 1: Horizontal Elastic Pendulum

A horizontal elastic pendulum is formed from a solid (S), of mass $m = 0.2$ kg, and a spring (R) of unjointed turns and of stiffness $K = 50$ N/m. When the spring takes its free length its center of inertia G of (S) is confounded with the center O of an axis $x'Ox$ parallel to the spring.



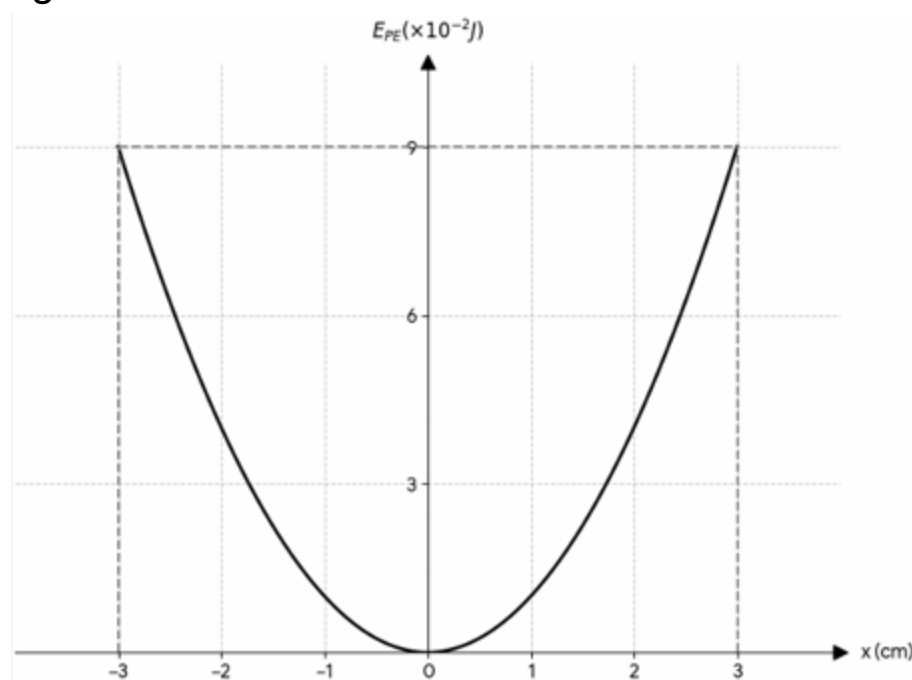
At the time $t_0 = 0$, we launch (S), which is found initially at O , with a velocity $\vec{V}_0$, of magnitude $V_0 = 2$ m/s, parallel to $x'Ox$ as shown in the figure. After launching (S) enters into free undamped oscillations. At time t , G is referred to by abscissa: $x = \overline{OG}$.

The reference level for the gravitational potential energy is the horizontal plane passing through O .

- Write the expression of mechanical energy of system (Pendulum ; Earth) at a time t and calculate its value.
- Determine the differential equation which governs the oscillations of (S).
 - Specify the nature of oscillations.
- Find the period of these oscillations. What does this period represent?
- The solution of the differential equation of the oscillations is under the form: $x = X_m \sin(\omega t + \varphi)$. Calculate X_m , ω , and φ .
- Represent the graph of x as a function of time.

Ex 2: Potential Energy of a Mechanical Oscillator

The graph of the potential energy of a free un-damped mechanical oscillator, in rectilinear motion on an axis $x'Ox$, in the interval $[-3$ cm; $+3$ cm], as a function of its elongation x is given in the figure below. The mass of the oscillator is $M = 400$ g.

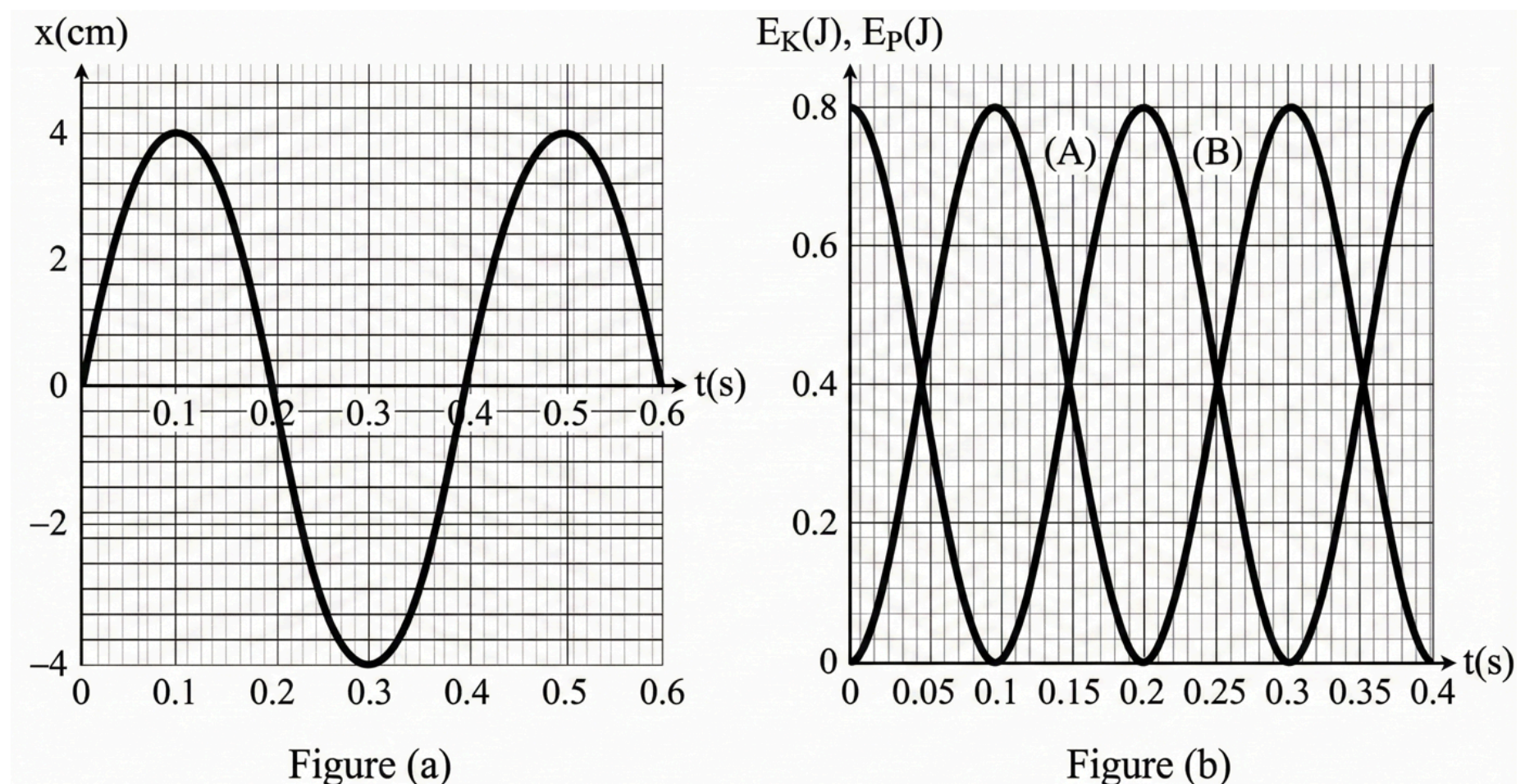


- Knowing that the potential energy of the oscillator is a parabolic function $E_p = ax^2$ where a is a constant.
 - Calculate a and give its unit.
 - Write the expression of the mechanical energy of the oscillator as a function of the elongation x and its speed x' .
- Determine the differential equation of the motion. Deduce the period of the oscillations.

3. Knowing that at $t = 0$, $x = -1$ cm and $x' < 0$.
 - a) Give the solution of the differential equation.
 - b) Represent the graph of x as a function of t .

Ex 3: Analysis of Elongation and Energy Graphs

The graph of the elongation x of the center of inertia of an elastic horizontal pendulum of mass M and stiffness K , as a function of time is given in figure (a). In figure (b) we represent the graphs of: the elastic potential energy and the kinetic energy of the pendulum as a function of time.



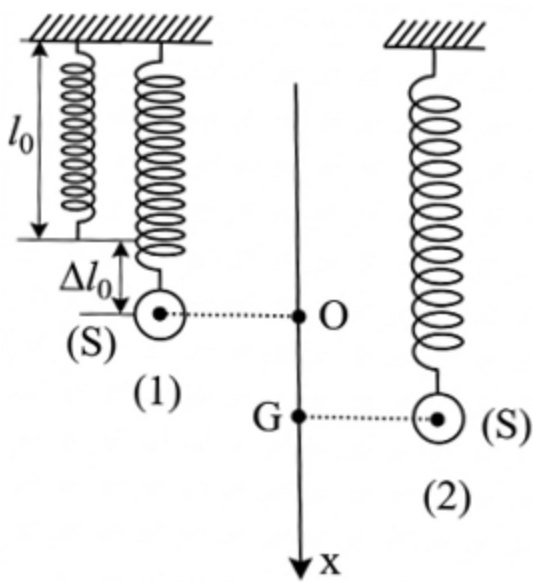
1. Calculate the frequency of the oscillator.
2. Associate, with justification, to the graphs (A) and (B) the corresponding energy.
3. a) Find the equation of x as a function of t . Deduce the maximum speed of the oscillator.
b) Calculate M and K .
c) Determine, by the aid of the graphs in figures (a) and (b), at the time $t = 0.24$ s, the position and the speed of the oscillator then verify these values using the equations of x and V as a function of t .
4. The curves (A) and (B) are periodic of the same period. Extract a relation that exists between the period of oscillations and the period of the kinetic or the potential energy.

Ex 4: Vertical Elastic Pendulum

We consider an elastic vertical pendulum, of stiffness k and whose solid (S) has a mass $m = 50$ g. The position of the center of inertia of (S) is referred to by G on a vertical descending axis $x'Ox$. At equilibrium G has zero abscissa (figure 1). We displace (S) vertically downward by 4 cm, from its equilibrium position, and we launch it without speed at time $t_0 = 0$.

The solid (S) enters into oscillation and has, at a time t , an abscissa $x = \overline{OG}$ (figure (2)) and a speed $V = \frac{dx}{dt}$.

The horizontal plane passing through O is taken as a reference level for gravitational potential energy. Take $g = 10$ m/s².



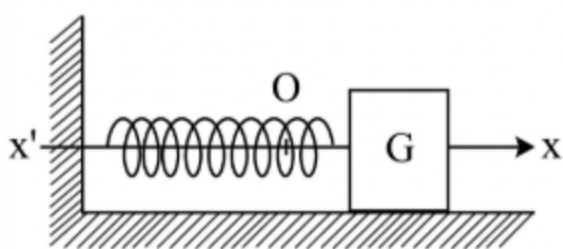
1. a) Calculate, as a function of m , g and k , the elongation $\Delta\ell_0$ of the spring at equilibrium.
b) Verify that the mechanical energy of the system (Earth ; Pendulum), at the time t , is:

$$E_m = \frac{1}{2}mV^2 + \frac{1}{2}kx^2 + \frac{1}{2}k\Delta\ell_0^2$$

2. We neglect friction.
 - a) The solid (S) passes by its equilibrium position with a speed $V_0 = 0.5 \text{ m/s}$. Applying the conservation of mechanical energy, verify that k is about 7.8 N/m .
 - b) Determine the differential equation which governs the oscillation of (S). c) Specify the nature of the oscillations and calculate their period.
3. We attach to (S) a metallic plate (P), thin, horizontal and of negligible mass. We displace (S) by 4 cm downward, from its equilibrium position at O, and we release it without speed, we find that it makes 8 oscillations before it stops at its equilibrium position.
 - a) Precise the type of oscillations.
 - b) Calculate the average power that must be supplied to the pendulum in order to maintain the oscillations.

Ex 5: Horizontal Oscillator (Damped vs Undamped)

A horizontal elastic pendulum oscillates freely around a horizontal and rectilinear support. The position of the center of inertia is referred to by $x = \overline{OG}$.



- The spring takes its initial length back when the center of inertia G is at O .
- The stiffness of the spring is $K = 85 \text{ N/m}$.
- The gravitational potential energy is zero on $x'Ox$.
- The graphs of x as a function of time in the two situations are represented in the figures below.

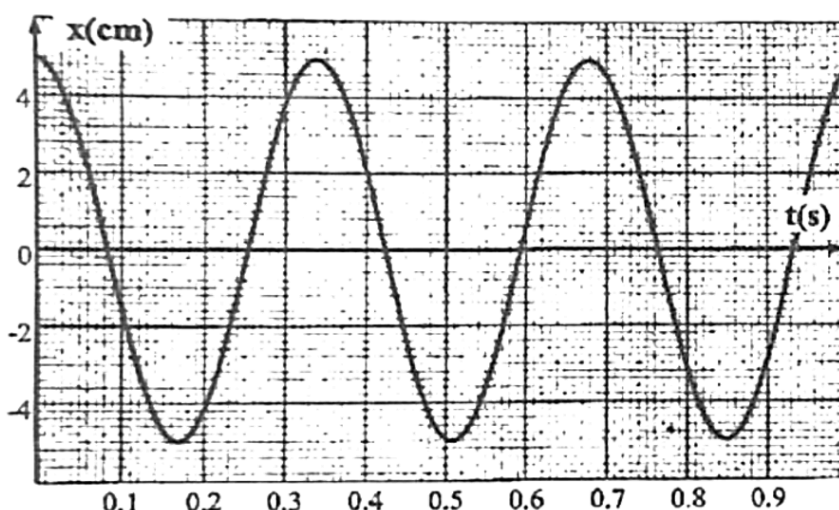


Figure (a)

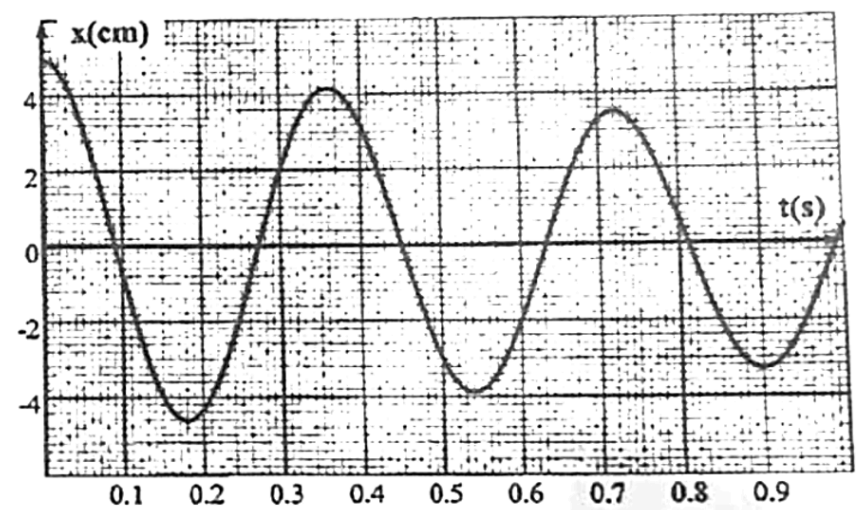


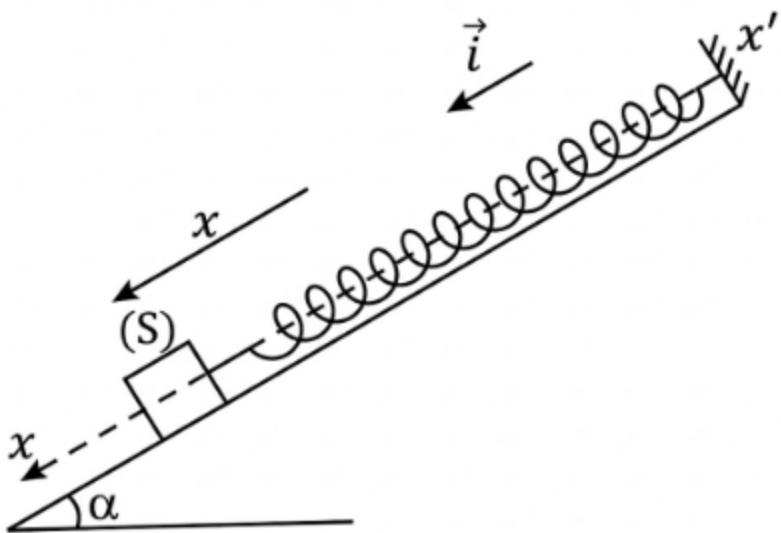
Figure (b)

1. Extract, in each of the above situations, the initial conditions (speed and x at $t = 0$) of the oscillations.
2. **Studying the oscillations in figure (a)**
 - a) The oscillations are free and un-damped. Justify.
 - b) Determine the differential equation described by these oscillations.
 - c) Deduce the expression of the period T of the oscillations. What is this period called?
 - d) Calculate the mass of the oscillator.
3. **Studying the oscillations in figure (b)**
 - a) The oscillations in figure (b) are damped. Justify.
 - b) What do we call the duration T' separating the passage of the abscissa x by two successive maxima?
 - c) Calculate the energy lost by the system (Earth ; pendulum) in the interval $[0; T']$.
 - d) Deduce the average power that should be given to the pendulum to support these oscillations.
 - e) Give the value the period of the driven oscillations.

Ex 6: Oscillator on an Inclined Plane

A spring, of negligible mass, of free length $\ell_0 = 20$ cm, of stiffness k , of unjointed turns, holds by one of its extremities a solid (S), of mass $m = 100$ g, and whose other extremity is fixed by a support.

The solid (S) is placed on a very smooth plane, inclined by an angle $\alpha = 30^\circ$ with respect to the horizontal. At equilibrium, the center of inertia of (S) is confounded with a point O and the length of the spring is $\ell = 25$ cm. Take: $g = 10$ m/s².



1. Calculate k .
2. Let $x'Ox$ be an axis, of unit vector $\vec{i}$, parallel to the spring and to the line of greatest slope of the inclined plane. We displace (S), along $x'Ox$, downward by a distance $d = x_0 = 5$ cm. At the time $t_0 = 0$, we launch (S) with a speed $\vec{V} = V_0\vec{i}$, it enters into sinusoidal oscillation, along the axis $x'Ox$, with an amplitude: $X_m = 8$ cm. At the time t the center of inertia G of (S) is referred to by the abscissa $x = \overline{OG}$. The reference level of gravitational potential energy is the horizontal plane passing through O .
 - a) Show that the mechanical energy of the system ((S) – spring – Earth), at time t , is:

$$E_m = \frac{1}{2}mV^2 + \frac{1}{2}kx^2 + \frac{1}{2}k\Delta\ell_0^2, \text{ with } \Delta\ell_0 = \ell - \ell_0$$

- b) Applying conservation of mechanical energy verify that:

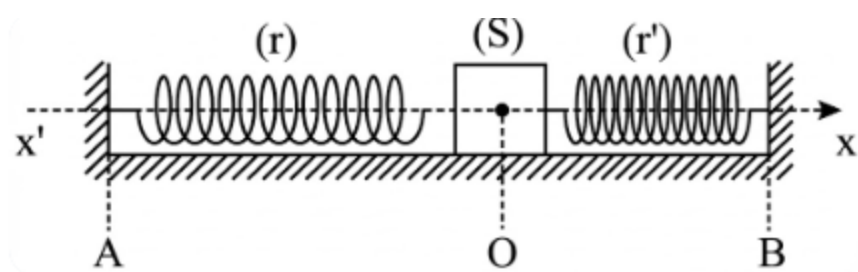
$$V = \pm \sqrt{\frac{g \sin \alpha}{\Delta\ell_0} (X_m^2 - x^2)}$$

3. a) Calculate V_0 .
b) Specify the position of (S) where its speed is maximum. Calculate this speed.
c) Determine the differential equation which governs the oscillation of (S) . Deduce the expression of the period of the oscillations.
4. The solution of the differential equation is under the form: $x = a \sin(b \cdot t + c)$.
a) Find the values of a , b and c (c has two possible values).
b) Represent the graph of x as a function of time.

Ex 7: Oscillator Between Two Springs

A solid (S) , of mass $m = 600$ g, can move without friction on a horizontal table. The solid (S) lies between two springs (r) and (r') , that are mass less, of the same axis $x'x$, of the same free length $\ell_0 = 20$ cm and of respective constants $k = 8$ N/m and $k' = 12$ N/m as shown in the adjacent figure.

At equilibrium, the position of the center of inertia G of (S) confounds with the center O of the axis $x'x$. Take $AB = 50$ cm and the gravitational potential energy to be zero at O .



1. Calculate, at equilibrium, the elongations $\Delta\ell_0$ and $\Delta\ell'_0$ of (r) and (r') .
2. Determine the potential energy of the system $((S) ; \text{springs} ; \text{Earth})$.
3. We displace (S) along $x'x$ so that G has an abscissa $x_0 = -4$ cm and we release it without speed at $t_0 = 0$.
a) Construct the figures at equilibrium and at a later time $t > 0$ where G has an abscissa $x > 0$.
b) Show that the mechanical energy of the system $((S) ; \text{springs} ; \text{Earth})$ has the expression:

$$E_m = \frac{1}{2} M v^2 + \frac{1}{2} (k + k') x^2 + 0.024$$

where x and v are respectively the abscissa and the speed of G .

- c) Find the value of this energy.
- d) Derive the differential equation of the motion of (S) and show that its frequency can be written as $\nu = \sqrt{f^2 + f'^2}$ where f and f' are two constants whose expressions are to be determined.
- e) Determine the solution of the differential equation. f) Deduce the speed of (S) at the instant $t = 2$ s.